**SECONDARY SCIENCE EXAMINATION- 2016**

**(ANNUAL)**

**SCIENCE**

**[TIME: 2 HOURS 30 minutes]** **F. M. 80**

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**Instructions for Candidates:**

1. **Candidates are required to give their answers in their own words as far as possible.**
2. **Figures in the right hand margin indicates full marks.**
3. **While answering the questions Candidate should adhere to the word limit as far as possible.**
4. **15 minutes of extra time has been allotted for the candidates to read the questions carefully.**
5. **This question paper is divided into two sections section A and section B.**

## SECTION – A

**1. How many laws are there for reflection of light? [1]**

Ans: There are three laws for reflection:

(1) The incident ray, Normal and reflected ray all lie in same plane.

(2) Incident ray and reflected ray always lie in opposite to the Normal

(3) Angle of incidence is always equal to angle of angle of reflection.

**2. What is Snell's law? [1]**

Ans: Snell's law is a formula used to describe the relationship between the angles of incidence and refraction, when referring to light or other waves passing through a boundary between two different isotropic media, such as water, glass, or air.

**3. What is the source of alternating current? [1]**

Ans: Hydroelectric power plants, AC generators produce alternating current.

**4. What is the mathematical form of Ohm's law? [1]**

Ans: The mathematical form of Ohm's law is:

$V=IR$

Where:

V = voltage expressed in Volts

I = current expressed in Amps

R = resistance expressed in Ohms

**5. What type of lens is used to correct nearsightedness? [1]**

Ans: Concave lenses

**6. What group does carbon belong to in the periodic table? [1]**

Ans: The carbon group is a periodic table group consisting of carbon (C), silicon (Si), germanium (Ge), tin (Sn), lead (Pb), and flerovium (Fl). In modern IUPAC notation, it is called Group 14.

**7. What is the atomic number of sodium? [1]**

Ans: The atomic number of Sodium is 11.

**8. What is the pH value of pure water? [1]**

Ans: The pH value of pure water is 7.

**9. What do the rows on the periodic table represent? [1]**

Ans: Each row of the periodic table represents a period.

**10. Write the name of the main alloys of aluminum metal? [1]**

Ans: Two main alloys of aluminium are:

1. Birmabright (aluminum, magnesium)
2. Duralumin (copper, aluminum)

**11. What is the digestion process in an amoeba? [1]**

Ans: Digestion in amoeba is intracellular taking place within the cell. The food taken in remains in a food vacuole or gastric vacuole formed by the cell membrane and small part of the cytoplasm. The vacuoles are transported deeper into the cells by cytoplasmic movements. Here they fuse with lysosomes that contain enzymes. Two enzymes amylase and proteinase have been reported. Thus, amoeba can digest sugars, cellulose and proteins. Fats, however, remain undigested.

The contents of the vacuole become lighter and the outline of the vacuole becomes indefinite indicating that the digestion is complete.

**12. What is the respiratory organ of cockroach? [1]**

Ans: The respiratory organ of cockroach is known as tracheae.

**13. Which organelle is called powerhouse of cell? [1]**

Ans: Mitochondria is called the powerhouse of cell.

**14. Which device is used to measure blood pressure? [1]**

Ans: A sphygmomanometer is used to measure blood pressure.

**15. Who is the father of genetics? [1]**

Ans: Gregor Mendel is the father of genetics.

**16. What is resistance? Write its S.I. unit. [2]**

Ans: Resistance is the opposition that a substance offers to the flow of electric current. It is represented by the uppercase letter R. The standard unit of resistance is the ohm.

**17. What is defect of vision? What are its types? [2]**

Ans: The inability of ciliary muscles to change the thickness of eye-lens as per requirement and to form the image of an object on retina leads to the defect of vision.

The three main common defects of vision are:

(i) Myopia or near-sightedness

(ii) Hypermetropia or far-sightednes

(iii) Presbyopia

**18. What is homogeneous mixture? Give example [2]**

Ans: Mixtures which have uniform composition throughout are called Homogeneous Mixture. For example – mixture of salt and water, mixture of sugar and water, air, lemonade, soda water, etc.

**19. How does the term ore differ from mineral give an example? [2]**

Ans: The difference between ore and mineral are:

| Mineral | Ore |
|---|---|
| 1. Naturally occurring substance of metals present in the earth's crust are called minerals. | 1. Minerals that can be used to obtain the metal profitably are called ores. |
| 2. All minerals are not ores. | 2. All ores are essentially minerals too. |
| 3. Clay is the mineral of Aluminium. | 3. Example: Bauxite and Cryolite are the main ores of aluminium. |

**20. Which disease is caused due to the deficiency of iodine and how? [2]**

Ans: Iodine deficiency may result in a goiter. A low amount of thyroxine (one of the two thyroid hormones) in the blood, due to lack of dietary iodine to make it, gives rise to high levels of thyroid stimulating hormone (TSH), which stimulates the thyroid gland to increase many biochemical processes; the cellular growth and proliferation can result in the characteristic swelling or hyperplasia of the thyroid gland, or goiter.

**21. How are gases exchanged in plants? [2]**

Ans: Plants obtain the gases they need through their leaves. They require oxygen for respiration and carbon dioxide for photosynthesis. The gases diffuse into the intercellular spaces of the leaf through pores, which are normally on the underside of the leaf - stomata. From these spaces they will diffuse into the cells that require them.

**22. What is the color reflection of light? How the spectrum is formed? [3]**

Ans: When light strikes a surface, some of its energy is reflected and some is absorbed. The color a person perceives indicates the wavelength of light being reflected. White light contains all the wavelengths of the visible spectrum, so when the color white is being reflected, that means all wavelengths are being reflected and none of them absorbed, making white the most reflective color.

Spectrum is formed of the light when it passes through a prism, light of different wavelength scatters according to Rayleigh scattering where' the amount a light is deflected is inversely proportional to the wavelength to the power of four'. Actually a white light consists of combination of different wavelength light VIBGYOR forming a single white light hence when they combine.



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**23. What is refraction of light? Mention its laws. [3]**

Ans: Refraction is the bending of a light ray as it passes at an angle from one transparent medium to another. As a beam of light enters glass at an angle, it is refracted or bent. The part of the light beam that strikes the glass is slowed down, causing the entire beam to bend. The more sharply the beam bends, the more it is slowed down. Each color has a different wavelength, and it bends differently from all other colors. Short wavelengths are slowed more sharply upon entering glass from air than are long wavelengths. Red light has the longest wavelength and is bent the least. Violet light has the shortest wavelength and is bent the most. Thus violet light travels more slowly through glass than does any other color.

Laws of Refraction

The angle of incidence is the angle between the incident ray and the normal; denoted as ‘i’. The angle of refraction is the angle between the refracted ray and the normal; denoted as ‘r’. Laws of refraction state that:

(a) The incident ray, reflected ray and the normal, to the interface of any two given mediums; all lie in the same plane.

(b) The ratio of the sine of the angle of incidence and sine of the angle of refraction is constant.

**24. Silver, gold, and platinum are used to make ornaments. Why? [3]**

Ans: Platinum, gold and silver are used to make jewellery because these are low reactive metals so rarely corrosion occurs and hence they do not lose their shine and luster.

**25. What is neutralization reaction? Give two examples [3]**

Ans: Neutralization is a chemical reaction in which acid and base combine to form salt and water. Acids donate $H^+$ and bases donate $OH^-$ in water which combine to form water molecules.

Example:

$HCL + NaOH \Longrightarrow NaCl + H_2O$

$CH_3COOH + NH_4OH \Longrightarrow CH_3COONH_4 + H_2O$

**26. What is bile? What is its role in human digestion? [3]**

Ans: Bile is a complex fluid containing water, electrolytes and a battery of organic molecules including bile acids, cholesterol, phospholipids and bilirubin that flows through the biliary tract into the small intestine.

Bile acids are derivatives of cholesterol synthesized in the hepatocyte. Cholesterol, ingested as part of the diet or derived from hepatic synthesis is converted into the bile acids cholic and chenodeoxycholic acids, which are then conjugated to an amino acid (glycine or taurine) to yield the conjugated form that is actively secreted into cannaliculi.

Bile acids are facial amphipathic, that is, they contain both hydrophobic (lipid soluble) and polar (hydrophilic) faces. The cholesterol-derived portion of a bile acid has one face that is hydrophobic (that with methyl groups) and one that is hydrophilic (that with the hydroxyl groups); the amino acid conjugate is polar and hydrophilic.

Their amphipathic nature enables bile acids to carry out two important functions:

(1) Emulsification of lipid aggregates: Bile acids have detergent action on particles of dietary fat which causes fat globules to break down or be emulsified into minute, microscopic droplets. Emulsification is not digestion per se, but is of importance because it greatly increases the surface area of fat, making it available for digestion by lipases, which cannot access the inside of lipid droplets.

(2) Solubilization and transport of lipids in an aqueous environment: Bile acids are lipid carriers and are able to solubilize many lipids by forming micelles - aggregates of lipids such as fatty acids, cholesterol and monoglycerides - that remain suspended in water. Bile acids are also critical for transport and absorption of the fat-soluble vitamins.

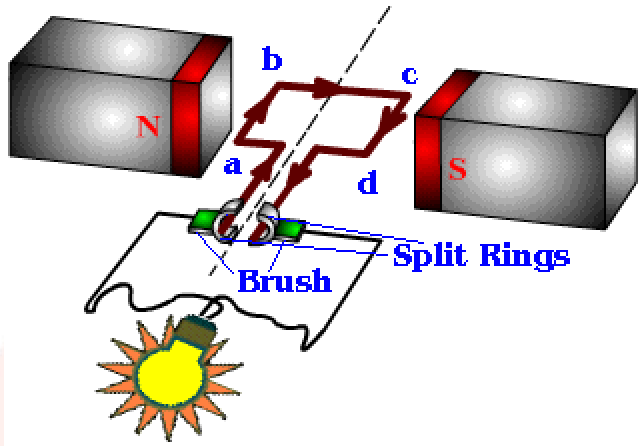
**27. What is excretory system? Write the name of its two main organs in humans. [3]**

Ans: Every living organism generates waste in its body and has a mechanism to expel it. In humans, waste generation and disposal are taken care of by the human excretory system. The human excretory system comprises of the following structures:

- 2 Kidneys
- 2 Ureters
- 1 Urinary bladder
- 1 Urethra

**28. What is dynamo? Illustrate its working principle with diagram. [5]**

Ans: Dynamo is an electric generator. It produces direct current by converting mechanical energy into electrical energy. It consists of a strong electromagnet between the poles of which is rotated an armature, consisting of a number of coils suitably wound. It is based on the principle of electromagnetic induction.




Working Principle of A DC Generator:

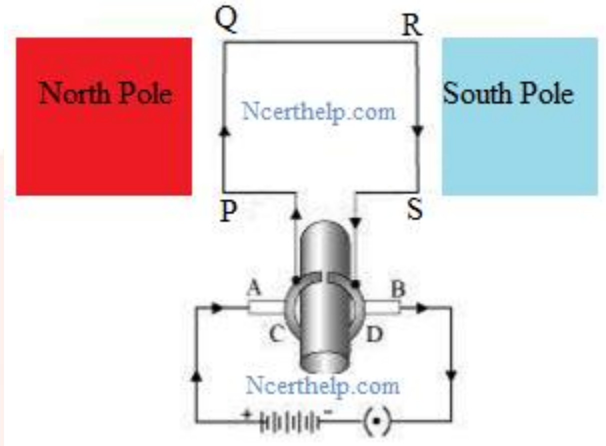
According to Faraday's laws of electromagnetic induction, whenever a conductor is placed in a varying magnetic field (OR a conductor is moved in a magnetic field), an emf (electromotive force) gets induced in the conductor. The magnitude of induced emf can be calculated from the emf equation of dc generator. If the conductor is provided with a closed path, the induced current will circulate within the path. In a DC generator, field coils produce an electromagnetic field and the armature conductors are rotated into the field. Thus, an electromagnetically induced emf is generated in the armature conductors. The direction of induced current is given by Fleming's right hand rule.

According to Fleming's right hand rule, the direction of induced current changes whenever the direction of motion of the conductor changes. Let's consider an armature rotating clockwise and a conductor at the left is moving upward. When the armature completes a half rotation, the direction of motion of that particular conductor will be reversed to downward. Hence, the direction of current in every armature conductor will be alternating. If you look at the above figure, you will know how the direction of the induced current is alternating in an armature conductor. But with a split ring commutator, connections of the armature conductors also gets reversed when the current reversal occurs. And therefore, we get unidirectional current at the terminals.

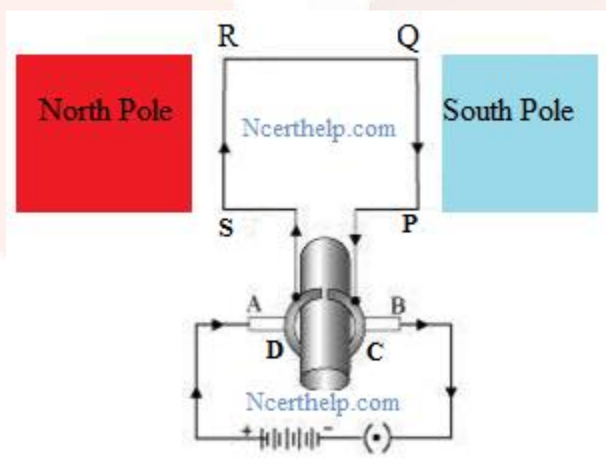
**OR**

**What is electric motor? Illustrate its theory and method of action with diagram. [5]**

Ans: An electric motor is a device which converts electrical energy into mechanical energy. The principle behind the electric motor is based on Fleming’s left hand rule.




Working: When a current is allowed to flow through the coil PQRS the coil starts rotating anti-clockwise. This happens because a downward force acts on length PQ and at the same time, an upward force acts on length RS. As a result, the coil rotates anti-clockwise. Current in the length PQ flows from P to Q and the magnetic field acts from left to right, normal to length PQ. Therefore, according to Fleming’s left hand rule, a downward force acts on the length PQ. Similarly at an upward force acts on the length RS. These two forces cause the coil to rotate anti-clockwise. When coil complete half rotation, the position of PQ and RS interchange. The half-ring D comes in contact with brush A and half-ring C comes in contact with brush B. therefore the direction of current in the coil PQRS gets reversed. Now the current flows through the coil in the direction SRQP.



The direction of current through the coil PQRS reverse after every half rotation. As a result, the coil rotates in same direction. The split rings help to reverse the direction of current in the circuit. These are called the commutator.

**29. What is ore? State the methods of concentration of ore. [5]**

Ans: An ore is a body of rock which contains a high enough concentration of a metallic or non-metallic miner or resource that the material can be profitably queried or mined, refined or otherwise processed, to separate the 'good stuff' and that 'good stuff shipped to a buyer and sold. The process of removal of the gangue from Ore is known as Concentration or Dressing or Benefaction. There are numerous methods of concentration and the methods are chosen based on the properties of the ore.

1) Hydraulic Washing: This method concentrates the ore by passing it through an upward stream of water whereby all the lighter particles of gangue are separated from the heavier metal ore. This is a type of gravity separation.
2) Magnetic Separation: This involves the use of magnetic properties of either the ore or the gangue to separate them. The ore is first ground to fine pieces and then passed on a conveyor belt passing over a magnetic roller. The magnetic ore remains on the belt and the gangue falls off the belt.
3) Froth Flotation Method: This method is mainly used to remove gangue from sulfide ores. The ore is powdered and a suspension is created in water. To this are added, Collectors and Froth Stabilizers. Collectors (pine oils, fatty acids etc) increase the non-wettability of the metal part of the ore and allows it to form a froth. Froth Stabilizers (cresols, aniline etc) sustain the froth. The oil wets the metal and the water wets the gangue. Paddles and air constantly stir up the suspension to create the froth. This frothy metal is skimmed off the top and dried, to recover the metal.
4) Leaching: Leaching is used when the ore is soluble in a solvent. The powdered ore is dissolved in a chemical, usually a strong solution of NaOH. The chemical solution dissolves the metal in the ore and it can be extracted and separated from the gangue by extracting the chemical solution. Extraction of the Aluminium metal from Bauxite ore is done using this process.

**OR**

**What is the difference between soap and detergent? [5]**

Ans: The most basic difference between soap and detergents is the ingredients. Soap is made primarily from some type of natural fat and lye. Detergents, on the other hand, are made from petroleum products and generally contain preservatives and antibacterial agents.

The other differences are:



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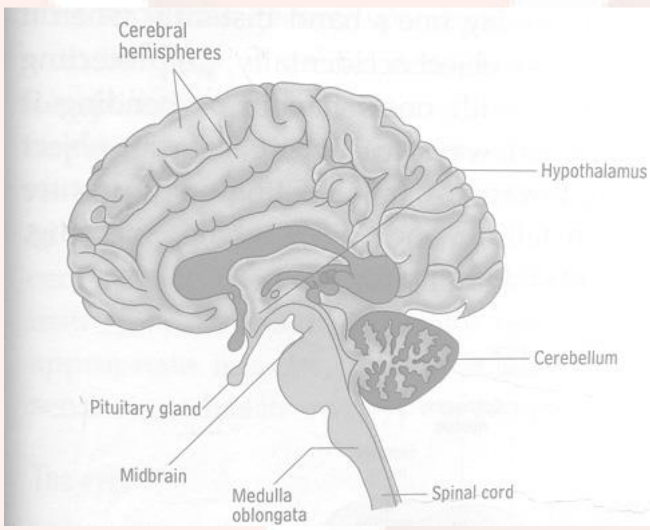
| Soaps | Detergents |
| --- | --- |
| 1) Soaps are sodium or potassium salts of long chain carboxylic acids.<br>2) Soaps have lesser cleansing action or quality as compared to detergents.<br>3) Soaps are made from animal or plant fats.<br>4) Soaps are more biodegradable. | 1) Detergents are ammonium or solphonate salts of long chain carboxylic acids.<br>2) Detergents have better cleansing action as compared to soaps.<br>3) Detergents are made from petrochemicals. Detergents are less biodegradable. |

**30. Draw a neat diagram of human brain and label [5]**

Ans:




**OR**

**Draw a neat diagram of Human Nephron and label [5]**

Ans:



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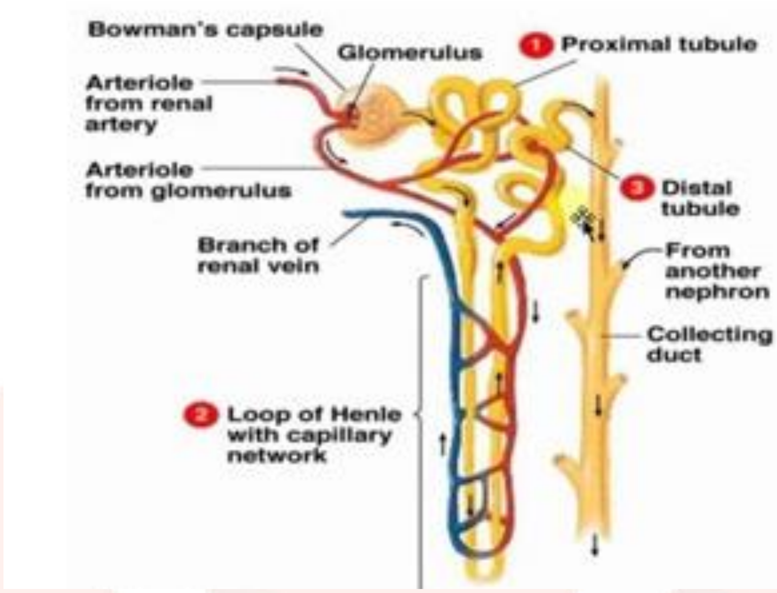


## SECTION B

**i) On which principle of electric current does electric fuse work:**

**a. Heating**
**b. Magnetic**
**c. Chemical**
**d. None of these**

Ans: Heating

**ii) The formula for determining resistance is:**

**a. R = VxI**
**b. R = V/I**
**c. R = I/V**
**d. R=V-I**

Ans: R = V/I

**iii) Which type of current flows through the battery:**

**a. Direct**
**b. Alternate**
**c. Both**
**d. None of these**

Ans: Direct

**iv) Which type of mirror is used for shaving?**

- **a. Concave**
- **b. Convex**
- **c. Plane**
- **d. None of these**

Ans: Concave

**v) What is the SI unit of potential difference:**

- **a. Volt**
- **b. Ampere**
- **c. Watt**
- **d. Coulomb**

Ans: Volt

**vi) How many laws of refraction are there:**

- **a. 1**
- **b. 2**
- **c. 3**
- **d. 4**

Ans: 3

**vii) 1 volt is called**

- **a. 1 joule per second**
- **b. 1 joule per coulomb**
- **c. 1 Joule per ampere**
- **d. None of these**

Ans: 1 joule per coulomb

**viii) Which type of chemical reaction is respiration?**

- **a. Oxidation**
- **b. Combination**
- **c. Exothermic**
- **d. Endothermic**

Ans: Exothermic

**ix) Which is the best conductor of electricity:**

- **a. Cu**
- **b. Ag**

c. **Al**
d. **Fe**

Ans: Ag

**x) In periodic table the vertical columns are known as:**

a. **Class**
b. **Period**
c. **Allotropes**
d. **None of these**

Ans: Class

**xi) Which of the following is a metalloid?**

a. **Fe**
b. **Cu**
c. **Ni**
d. **Cb**

Ans: Cb

**xii) Brass is:**

a. **Metal**
b. **Nonmetal**
c. **Alloy**
d. **Metalloid**

Ans: Alloy

**xiii) pH value of an acidic solution is:**

a. **7**
b. **Less than 7**
c. **More than 7**
d. **None of these**

Ans: Less than 7

**xiv) The simplest hydrocarbon is**

a. **Methane**
b. **Ethane**
c. **Propane**
d. **Butane**

Ans: Methane

**xv) Pseudopodia is found in**

a. **Paramecium**
b. **Euglena**
c. **Amobea**
d. **None of these**

Ans: Amoeba

**xvi) In flowering plant, sexual reproduction is done by:**

a. **Leaves**
b. **Flowers**
c. **Stem**
d. **None of these**

Ans: Flowers

**xvii) Which of the following is an excretory organ:**

a. **Kidney**
b. **Pancreas**
c. **Eyes**
d. **None of these**

Ans: Kidney

**xviii) Sperm is formed in**

a. **Testis**
b. **Urethra**
c. **Uterus**
d. **Ovum**

Ans: Kidney

**xix) Which of the following is not a natural resource**

a. **Air**
b. **Soil**
c. **Water**
d. **Living organism**

Ans: Living Organism

xx) **Which of the following is caused due to lack of insulin?**

  a. **Goiter**
  b. **Dwarfism**
  c. **Diabetes**
  d. **None of these**

Ans: Living Organism



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